

Seokmin Shin

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Integrated MS/PhD student

Department of Material Science & Engineering

Seoul National University

EDUCATION

- **Seoul National University, SNU** September 2026 - Current
Integrated MS/PhD, Materials Science & Engineering Seoul, South Korea
 - Advisor: **Prof. Stephen Dongmin Kang**
- **Ulsan National Institute of Science and Technology, UNIST** March 2018 - June 2024
Integrated MS/PhD Ulsan, South Korea
 - (Completed Qualifying Exams & Coursework; Dissertation Research in Electrocatalysis)
- **Ulsan National Institute of Science and Technology, UNIST** March 2014 - March 2018
Undergraduate Ulsan, South Korea

ACADEMIC TRANSITION NOTE

My doctoral journey at UNIST led to several peer-reviewed publications, conference presentations, patents, and awards in the field of electrocatalysis, centered on engineering better catalysts through empirical screening and the circumventing the linear scaling relationships (LSRs). During the final stage of the program, exposure to frontier research at the 2024 Gordon Research Conference on Electrochemistry, together with personal caregiving responsibilities that prompted deeper reflection, brought a different question into focus. "From how to engineer a better electrode/catalytic platform, to how to define what is measured with sufficient precision to be expressed as a separable variable within a rigorous thermodynamic framework". That question shaped my decision to join Prof. Stephen Dongmin Kang's group at Seoul National University, where state-pinned measurement architectures and state-resolved electrochemical kinetics continue my earlier inquiry.

RESEARCH EXPERIENCE

- **Deciphering electrochemical kinetics of Li intercalation active materials** March 2026 - Current
Assistant Researcher, SNU Advisor: Stephen Dongmin Kang
 - * Design of state-fixed measurement architectures to pin surface state and isolate intrinsic charge-transfer kinetics from architecture-contaminated apparent parameters
 - * Co-authored a measurement-methodology paper on the challenges of resolving intercalation charge-transfer kinetics (ACS Energy Letters, 2026)
- **Fundamental and engineering studies in electrochemical multi-reactant catalysis** March 2023 - June 2024
PhD Candidate, UNIST PI: Youngkook Kwon
 - * Understanding the catalytic mechanism under competing reactants
 - * Development of electrocatalyst for electrochemical co-reduction of carbon dioxide/nitrate
- **Controlling intermediate binding energy through surface-confined catalysts** March 2018 - March 2023
PhD Candidate, UNIST Advisor: Hyun-Kon Song
 - * Modulated the energetic linear scaling relationship (LSR) of key intermediates by experimental demonstration of atomic scale spacings into metal oxide based electrocatalysts utilizing lithiation/de-lithiation method
 - * Controlling the selectivity between oxygen and chlorine during seawater splitting
 - * Enhanced electrochemical performance via associative activation through organic/inorganic hybrid catalyst system (Nitrogen oxygenates conversion, oxygen reduction) | Altered proton transfer step in catalytic system
 - * Enzymatic CO₂ reduction to formate utilizing polymeric mediator and enzyme via electrode immobilization
- **Development of facile synthetic method for efficient oxygen electrocatalyst** September 2016 - March 2018
Undergraduate Student Researcher Advisor: Hyun-Kon Song
 - * The design and synthesis of efficient bi-functional electrocatalyst for oxygen evolution and reduction (Ni, Co, Fe, and Pt based electrocatalysts)

RESEARCH SKILLS

- **Electrochemical Techniques:** Galvanostatic and potentiostatic measurements | Electrochemical impedance spectroscopy | Coin-cell and state-fixed cell fabrication and testing | Cyclic voltammetry, linear sweep voltammetry, and chrono-potential/amperometry | Tafel and mass transport analysis
- **Prior Experimental Techniques (2018-2024, Electrocatalysis):** Wet/solid-state synthesis, X-ray-based operando analysis (XAFS, STXM), electron microscopy, and standard surface/compositional characterization (XPS, BET, ICP-OES/MS, Raman, NMR, GC/LC/IC)

RESEARCH INTEREST

• State-Resolved Electrochemical Kinetics

Measurability-first approach to charge transfer under (electro-)chemical regimes

- Investigating mechanistic frameworks that bridge electrochemical kinetics (e.g., Butler-Volmer, Marcus theory) with chemical free energy landscapes, grounded in a Gibbs–Guggenheim viewpoint
- Designing state-fixing measurement architectures to pin surface stoichiometry and separate intrinsic from apparent (architecture-contaminated) kinetic parameters
- Applying chemical potential-based frameworks to interpret charge transfer kinetics of intercalation materials

• Historical and Cross-Field Framing of Measurement Traditions

Positioning measurement architectures as tools that answer mechanistic questions across traditions

- Mapping the divergence between the operationalist/measurability tradition (Nernst, Gibbs, Guggenheim) and the microscopic mechanism tradition (Butler, Marcus, Hush) in electrochemical kinetics
- Applying measurability-tradition tools to mechanism-tradition questions, structurally paralleling approaches used in electrocatalytic charge-transfer decomposition
- Drawing on prior electrocatalysis background (linear scaling relationship-breaking, mechanism-vs-performance framing) to bridge battery and catalysis kinetics literatures

• Mechanistic Electrochemistry and Analytical System Design

Deconvolution of kinetic factors for mechanism study using electrochemical analysis

- Fundamental understanding of electrochemical interphases (phase boundaries of system)
- Designing cell systems for the analytical deconvolution of kinetic factors during reaction
- Deconvolution of mechanistic factors from observed kinetics through advanced electro-analysis

• Operando and *in-situ* Analysis for Mechanistic Investigation

Mechanism study utilizing various operando and in-situ analysis

- Study of charge transfer dynamics and structural evolution using X-ray based techniques (XAFS, STXM)
- Study of state-resolved observation of charge transfer
- Development of in-situ/operando analytical system for dynamic analysis

PUBLICATIONS

F=FIRST-AUTHORED, C=CO-AUTHORED

- [C.0] Y. Choi, H. Kweon, **Seokmin Shin**, S. D. Kang, “Charge-Transfer Kinetics of Intercalation Reactions: A Measurement Challenge”, **ACS Energy Letters** (2026)
- [F.1] **Seokmin Shin**, S. Sultan, Z.-X. Chen, H. Lee, H. Choi, T.-U. Wi, C. Park, T. Kim, C. Lee, J. Jeong, H. Shin, T.-H. Kim, H. Ju, H. C. Yoon, H.-K. Song*, H.-W. Lee*, M.-J. Cheng*, Y. Kwon*, “Copper With an Atomic-scale Spacing for Efficient Electrocatalytic Co-reduction of Carbon Dioxide and Nitrate to Urea”, **Energy and Environmental Science**, **16**, 2003-2013 (2023) Featured as Inside Back Cover
- [F.2] **Seokmin Shin**, T.-U. Wi, T.-H. Kong, C. Park, H. Lee, J. Jeong, E. Lee, S. Yoon, T.-H. Kim, H.-W. Lee, Y. Kwon*, H.-K. Song*, “Selectively Enhanced Electrocatalytic Oxygen Evolution within Nanoscopic Channels Fitting a Specific Reaction Intermediate for Seawater Splitting”, **Small**, **19**, 2206918 (2023)
- [C.1] H. Lee, N. Park, T.-H. Kong, S. Kwon, **Seokmin Shin**, S. G. Cha, E. Lee, J. Cha, S. Sultan, Y. Kwon*, “Advancements in electrochemical methanol synthesis from CO₂: Mechanisms and catalyst developments”, **Nano Energy**, **130**, 110099 (2024)
- [C.2] S. Kwon, T.-H. Kong, N. Park, P. Thangavel, H. Lee, **Seokmin Shin**, J. Cha, Y. Kwon*, “Directions of Oxygen Evolution Reaction Electrocatalyst Evaluation for Anion Exchange Membrane CO₂ Electrolyzer”, **EES Catalysis**, **2**, 911-922 (2024)

- [C.3] T.-H. Kong, P. Thangavel, **Seokmin Shin**, S. Kwon, H. Choi, H. Lee, N. Park, J.-J. Woo, Y. Kwon*, "In-situ Ionomer-free Catalyst Coated Membranes for Anion Exchange Membrane Water Electrolyzers", **ACS Energy Letters**, **8**, 4666-4673 (2023)
- [C.4] Y. Lee, J. H. Ahn, **Seokmin Shin**, S.-H. Jung, H.-S. Park, Y.-G. Cho, D.-G. Lee, H. Kong, J. H. Lee*, H.-K. Song*, "Metal-nitrogen intimacy of the nitrogen-doped ruthenium oxide for facilitating electrochemical hydrogen production", **Applied Catalysis B: Environmental**, **303**, 120873 (2022)
- [C.5] D.-G. Lee, S. H. Kim, J. Lee, **Seokmin Shin**, S. H. Joo, Y. Lee, C. Park, Y. Kwon, S. K. Kwak*, H.-K. Song*, "Double activation of oxygen intermediates of oxygen reduction reaction by dual inorganic/organic hybrid electrocatalysts", **Nano Energy**, **86**, 106048 (2021)
- [C.6] D.-G. Lee, S. H. Kim, H. H. Lee, **Seokmin Shin**, J. Lee, S. H. Joo, Y. Lee, S. K. Kwak*, H.-K. Song*, "Breaking the Linear Scaling Relationship by a Proton Donor for Improving Electrocatalytic Oxygen Reduction Kinetics", **ACS Catalysis**, **11**, 12712-12720 (2021)

AWARDS AND HONORS

- **The great oral presentation award**, The Korean Electrochemistry Society Spring 2023
- **Outstanding graduate student award**, Ulsan National Institute of Science and Technology Winter 2023
- **Samsung Human-tech paper awards**, Samsung Advanced Institute of Technology Spring 2024

CONFERENCE PRESENTATIONS

1. **Seokmin Shin**, T.-U. Wi, M.-J. Cheng, H.-W. Lee*, and Y. Kwon*, "Electrochemical C-N Bond Formation from Carbon Dioxide and Nitrogen Oxygenates", Electrochemistry: Gordon Research Seminar/Conference, Jan. 2024 [Poster]
2. **Seokmin Shin**, Y. Kwon*, "Copper with an Atomic-Scale Spacing for Efficient Electrocatalytic Co-reduction of Carbon Dioxide and Nitrate to Urea", The Korean Electrochemical Society annual meeting, 2023 [Oral]
3. **Seokmin Shin**, Y. Kwon*, "Copper with an Atomic-Scale Spacing for Efficient Electrocatalytic Co-reduction of Carbon Dioxide and Nitrate to Urea", International Society of Electrochemistry, 74th annual, France, 2023 [Oral]
4. **Seokmin Shin**, Y. Kwon*, H.-K. Song*, "Atomic-Scale Spacings Controlling Intermediate Binding Energy for Selective Electrocatalysis", The Korean Electrochemical Society annual meeting, Spring, 2023 [Oral]
5. **Seokmin Shin**, T.-U. Wi, H.-W. Lee, Y. Kwon*, H.-K. Song*, "Selectively Enhanced Electrocatalytic Oxygen Evolution Within Nanoscopic Channels Fitting a Specific Reaction Intermediate for Seawater Splitting", The Korean Electrochemical Society annual meeting, Fall, 2022 [Oral]
6. **Seokmin Shin**, T.-U. Wi, H.-W. Lee, Y. Kwon*, H.-K. Song*, "Selectively Enhanced Electrocatalytic Oxygen Evolution Within Nanoscopic Channels Fitting a Specific Reaction Intermediate for Seawater Splitting", International Society of Electrochemistry and The Korean Electrochemical Society, 7th Ertl Symposium on Catalysis in Electrochemistry, 2022 [Poster]
7. **Seokmin Shin**, D.-G. Lee, Y. Kwon, H.-K. Song*, "Selective Ammonia Production via Associative Activation of Nitric Oxide (NO) Intermediates by Inorganic/organic Hybrid Electrocatalysts", The Korean Institute of Chemical Engineers annual meeting, Spring, 2022 [Oral]
8. **Seokmin Shin**, D.-G. Lee, S. Yoon, Y. Lee, H.-K. Song*, "Electrocatalytic Oxygen Reduction Reaction Improved by Facilitated Proton Transfer", Electrochemical Society Meeting Abstracts prime2020, **68**, 3632-3632
9. **Seokmin Shin**, D.-G. Lee, H.-K. Song*, "Electrocatalytic Oxygen Reduction Reaction Improved by Facilitated Proton Transfer", The Korean Electrochemical Society annual meeting, Fall, 2018 [Poster]

PATENTS

KR patent: Hyun-Kon Song, Youngkook Kwon, **Seokmin Shin**, "Oxygen Evolution Reaction Catalyst For Seawater Electrolysis And Preparation Method Thereof", (10-2022-0100538)

KR patent: Yeongdae Lee, Hyun-Kon Song, Dong-Gyu Lee, Jisu Lee, **Seokmin Shin**, "Apparatus For Generating Hydrogen Using Hydrogen Peroxide", (10-2022-0071800)

TEACHING AND RESEARCH ASSISTANCE EXPERIENCE

Prof. Hyun-Kon Song, Department of Energy and Chemical Engineering, UNIST

- **Introduction to Chemical Engineering**, Undergraduate course Spring 2018
- **Electrochemistry**, Undergraduate course Fall 2018
- **Advanced Electrochemistry**, Graduate Course Spring 2019
- **Electrochemistry**, Undergraduate Course Fall 2019

Discussion and review sessions | Graded assignments, exams, and reports

Provided office hours and 1:1 academic support | Assisted in-class demonstrations and set-ups